

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 18

Limit Theorems

Theorem: (Strong Law of Large Numbers) Let $\{X_n\}$ be a sequence of i.i.d. RVs with finite mean μ . Define $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Then $\{\bar{X}_n\}$ converges to μ almost surely.

Example 1: Bernoulli proportion converges to success probability.

Example 2: Monte Carlo Integration.

Theorem: (Central Limit Theorem) Let $\{X_n\}$ be a sequence of i.i.d. RVs with mean μ and variance $\sigma^2 < \infty$. Then, as $n \rightarrow \infty$,

$$P\left(\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq a\right) \rightarrow \Phi(a) = \int_{-\infty}^a \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt.$$

Example 3: $X_n \sim \text{Bin}(n, p)$. Then

$$P\left(\frac{X_n - np}{\sqrt{np(1-p)}} \leq a\right) \rightarrow \Phi(a).$$

Example 4: The lifetimes of a special type of battery is a RV with mean 40 hours and standard deviation 20 hours. A battery is used until it fails, at which point it is replaced by a new one. Assume a stockpile of 25 such batteries, the lifetimes of which are independent, approximate the probability that over 1100 hours of use can be obtained. [$\Phi(1) = 0.8413$]